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ARETE • Automated REsearch Transparency Ecosystem

Connected claims for self-correcting science.

"If I have seen further, it is by standing on the shoulders of giants." — Isaac Newton

What are you trying to do?

Scientific progress builds on a network of claims. But progress is only reliable if claims' chains of evidence are transparent, as new evidence might falsify earlier findings. Such updates need to be propagated through the scientific corpus, but science lacks the infrastructure and the protocol to track this network. AI-generated research is flooding science with more errors than before: a scalable way to automatically verify or correct the record is essential but missing. ARETE builds the tools and the open platform to standardize scientific claims into a network with their chains of evidence, including the data, the assumptions, and the literature. The network automatically updates in light of new evidence. Verification tools that leverage autoformalization certify whether claims are supported by their evidence and whether their experimental design allows them. The program is a success if, after three years, a pilot journal adopts the platform to represent its papers' claims.

How is it done today?

Newton's shoulders-of-giants argument describes an ideal of scientific progress: that the web of science, represented by references, reliably tracks the chain of evidence behind scientific claims. This was never true, as scientific findings were always locked into static prose, burdensome to update in light of new evidence. Updating is essential, for **science has never been perfect** [8, 9, 16, 26, 27]: honest mistakes, but also misconduct have slipped through peer review. Without a dynamic network of scientific claims, falsified claims can be built upon for a long time: e.g., even after a randomized controlled trial found that estrogen plus progestin raised the risk of dementia, an observational study's causal claim that estrogen protects against Alzheimer's—made without causal evidence—was still mostly cited favorably [32–35]. Without a standard way to represent experimental protocols and assumptions, we are risking repeating the cancer-biology replication project, where missing details made none of almost 200 experiments repeatable from the paper alone [13]. Without having verification tools that help scientists check whether their methodology supports, e.g., causal claims, we are risking overwhelming the web of science with unjustified claims [18].

The downside is already substantial: wasted billions in public money, years of research, and eroding trust in science [11, 15, 30]: $28 billion a year goes to irreproducible preclinical research in the United States alone [15].

Why should we act now?

More science is published than anyone can check, and with more errors than before, driven by AI-generated works with plausible-sounding false claims [3, 19, 41]. The yearly volume on the preprint server arXiv almost trebled between 2015 and 2025 [4, 6], and the AAAI conference's submissions more than doubled in the last year [10]. Human peer review already took over 100 million hours worldwide in 2020, costing over $1.5 billion in the US alone [1]. Even mathematics now warns that AI can produce "plausible but unreliable (or even incorrect) arguments", and asks for "insisting on formal verification when appropriate" [2]; a Nature editorial asked other fields to follow [24]. In the last 3 years, **the share of papers with a fabricated reference has risen 10x in a biomedical open-access corpus** [38]. Of the papers affected, 98.4% had received no publisher action at the time of the audit [38]. **There is no network and no protocol that would automatically propagate a correction to the claims built on it**. This is substantially affecting clinical practice, as many clinical guidelines rely on meta-analyses

that included retracted clinical trials—excluding those would reverse the direction of the pooled effect in about 8% of approximately 4,000 meta-analyses [40].

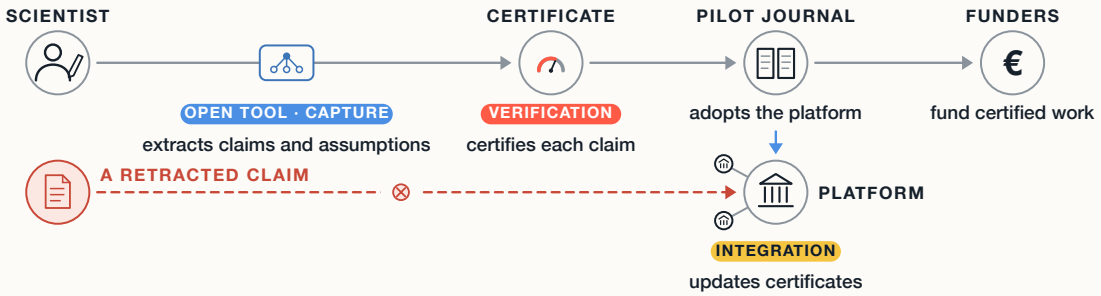
The technical challenge

Creating a scientific claim requires scientists to reason given their observations, experiments, data, and prior knowledge, and then to describe their often partially implicit reasoning and assumptions such that others can reproduce and build upon them. Building up the expertise to assess scientific claims takes years of training, rendering this verification step the bottleneck in the era of AI-assisted scientific discovery. The technical challenge to create the network infrastructure that represents scientific claims and their chain of evidence in a standardized, reliable, and open-access format is threefold:

1. **Scale:** the scientific corpus has over 320 million works [28], with multiple interrelated claims each
2. **Validity:** assumptions, design, or evidence might be implicit, missing, ambiguous, or confidential
3. **Trust:** the network needs a protocol to invalidate falsified claims and update the certificates built on them

What is new, why will it work, and who cares?

ARETE proposes an ecosystem that turns science into a network of standardized claims that can be checked and tracked by a machine, making the evidence each claim rests on transparent, and any course-correction, should one claim be falsified, automatic. A scientist submitting to a journal runs the open tool over the paper, corrects the claims and assumptions it extracted, and receives a certificate for each; the reviewers read them beside the paper, and funders fund certified work. When a claim the paper rests on is retracted or falsified, every dependent claim's certificate is updated and the scientists behind it are notified.



The environment and the technical conditions changed in recent years, removing the blockers earlier programs faced: **language models became cheap and prevalent**, providing the backbone to convert scientific claims, locked into natural language, into a standardized format. **Publishers and funding agencies became receptive to change**, as the peer review system cannot ensure the integrity of science anymore; they want a solution that reliably screens scientific claims [5, 7, 12, 14, 20, 25, 36, 37, 39].

What is **still missing** are the **open standard, certificate, protocol, and infrastructure** that represent these claims and their relationships, and track their provenance; the **extraction methods** that turn scientific claims formulated in natural language into a structured format with all assumptions and the design stated explicitly; and the **formal checkers** that tell whether the claim is supported and whether the design allows it.

LLMs can help with claim extraction and autoformalization, but they fall short in faithfully capturing the assumptions or in formally verifying conclusions [29, 31]. Thus, we need independent verification tools for scientific reasoning, to ensure the extraction methods correctly represent the scientific claims. This would require formalizing experimental design, building on and integrating tools like Lean for math and XDL for chemistry [17, 21, 22].

Although provenance exists at a global scale for cryptocurrencies [23], we still need to build one that keeps track of all scientific artifacts and experimental design in a standard format and automatically

propagates updates (such as new results or retractions) through the network. The resulting platform will enable automatic course-correction if a claim later turns out to be mistaken, stopping any ongoing effort (such as a clinical trial) building on that claim. Since the standard and certification are automated and machine-readable, AI agents can even use this ecosystem to test new hypotheses at scale on top of already checked claims.

Program design

To implement the missing pieces detailed above, this 36-month program proposes a modular ecosystem with independent validation, resulting in three Technical Areas:

- **Integration** is responsible for the standard claim format, the certificate, and the distributed platform that stores the claims and tracks where each piece of evidence came from, and the protocol that updates the claims' certificates.
- **Capture** develops methods to extract scientific claims and their assumptions and context into a standardized format developed by Integration.
- **Verification** develops the formal methods to verify the reasoning behind scientific claims, and evaluates the Capture teams' extraction methods on benchmarks, which it also creates and maintains.

The benchmarks encompass multiple scientific fields, focus on well-defined correlational/causal claims, and have multiple, increasingly difficult splits that the Capture teams cannot see during development.
Metrics. Capture's main metric is the graded overclaim, whether the extracted claims and their assumptions imply at most the right level of confidence, measured at a floor on the coverage of the claims, and at most 5% of the claims their evidence supports may be denied a certificate. Every denial comes with a counterexample: two models of the study that fit its design and evidence but disagree on the claim. A faithfulness audit, scored separately, tests whether the formalized claim still says what the paper said. A certificate holds relative to the assumptions the claim itself lists.
Stages and gates. In Stage 1, Verification and Integration start by developing the formal checkers with the public development split of the benchmark, and the claim standard, respectively. In Stage 2, Integration delivers the platform with provenance tracking and Verification the hidden splits; in Stage 3, the correction protocol and the planted false claims.

Capture starts after the release of the development split of the benchmark, and each team is evaluated at each gate against a tighter cap on the graded overclaim and across wider coverage, where overclaim is defined in terms of a formal hierarchy of correlational/causal claims; the quantitative Gates are in the table below.

	Teams	Per team	Scope	Coverage floor	Graded overclaim
Stage 1	8	up to €0.7M	a single paper	75% of its claims	15%
Stage 2	6	up to €1.7M	the paper and its references	95% of its claims, 75% of the relevant claims in cited works	10%
Stage 3	5	up to €1.6M	a three-deep network of references	95% of its claims, 90% of the relevant claims across the network	5% (stretch: under 2%)

Every gate also audits faithfulness, scored separately from the overclaim.

If every Capture team reaches the benchmark's ceiling, or the benchmark enables an unforeseen solution with a high metric but an unwanted mechanism, the program extends or pivots to additional benchmarks.
Deliverables. Stage 3 leaves a governed open standard, a reference implementation, the formal checkers released as open source, and the platform with the Capture teams' methods integrated into it. A consortium of publishers, funders and universities, an existing standards body, and an independent nonprofit on the arXiv model are the candidate homes after the program.
Cost and structure. The program has a single Integration team and two Verification teams in every stage. Verification teams build a common benchmark but separate formal checkers to improve robustness. Capture teams compete on the extraction methods: five start and narrow to three, then two, so that two independently built extraction pipelines reach the Stage-3 gate. The program costs €27.5M gross over three years. €22.8M funds the teams' grants, including the Verification teams' evaluation budget and a post-grant steward for the standard, €2.3M funds the shared tracks for independent judging, and €2.4M is contingency.

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Table 1 Each row names one person consulted on this program or who approached it, with their affiliation.

Name	Affiliation
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Hakeem Angulu	Anthropic, member of technical staff
Luca Moschella	Apple, machine-learning researcher
Alice Pettitt	ARIA
Andy Gordon	ARIA, science advisor; University of Edinburgh, honorary professor
Maximilian Sichrovsky	ARIA
Nora Ammann	ARIA, program director
Pranay Shah	ARIA, head of special projects
Kumar Garg	Renaissance Philanthropy
Max Bode	Biocompile, co-founder and chief product officer
Lukas Wallrich	Birkbeck Business School, University of London, senior lecturer
Tim Errington	Center for Open Science, chief research officer
Brian Nosek	Center for Open Science, co-founder and executive director
Eileen Clancy	Center for Open Science, managing editor of Lifecycle Journal
Shakhlo Nematova	Center for Open Science, research scientist
François Lanusse	CNRS, researcher; Lightcone Research, co-lead
Matt Clancy	Coefficient Giving, program director
Harry Dean Hudson	Community Computer (Radicle), contributor
Ronen Tamari	Cosmik, co-founder
Patrick Duffner	Cyber Valley, business development manager
Tim Vines	DataSeer, founder and chief executive
Witold Więcek	Development Innovation Lab, University of Chicago, consulting director
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Mike Dodds	Galois, principal scientist
Kirstie Weston	Google DeepMind
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Steffen Schneider	Helmholtz Munich, research group leader

Name	Affiliation
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Francesco Locatello	Institute of Science and Technology Austria, assistant professor
Valentin Rodionov	Intellicat, chief executive
Elizabeth Bismut	Inversion Lab, co-founder and managing director
Lisa Emme	Inversion Lab, co-founder and managing director
Janna Schweigmann	Inversion Lab, head of staff
Dan Elton	The Metascience Observatory, founder and director
Wolfgang Kerzendorf	Michigan State University, assistant professor
Simon Lacoste-Julien	Mila, associate scientific director and core academic member
Gaurav Sahu	Unified Sciences and Mila
Guy Wolf	Mila, core academic member
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Hugo Larochelle	Mila, scientific director
Michael Chiang	National Eye Institute, director
Richard Scheuermann	National Library of Medicine, scientific director
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Duncan Webb	Nova School of Business and Economics, assistant professor
Thomas Dietterich	Oregon State University, distinguished professor emeritus
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Craig Hughes	Reasonable AI, co-founder
Ralph Abboud	Renaissance Philanthropy
Fahd Husain	Renaissance Philanthropy, fellow
Natalie Khalil	Reviewer3, co-founder and chief executive
Paul Litvak	Robyn Dawes Institute, founder and executive director
Emanuele Rodolà	Sapienza University of Rome, full professor
Donato Crisostomi	Sapienza University of Rome, postdoctoral researcher
Anita Bandrowski	SciCrunch, founder and chief executive
Sami Benchekroun	SPRIND; Molecular Connections
Sebastian Berns	SPRIND, challenge officer
Barbara Diehl	SPRIND, director of IP Transfer 3.0
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Jamie Cummins	University of Bern, senior postdoctoral researcher
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Chenhao Tan	University of Chicago, associate professor
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